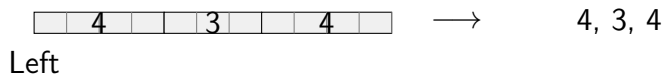


Build beside the page with whole rods of length 3 and 4. A word gives their lengths from the fixed left end to the right. Different orders count as different words. Rods of the same length are alike.



Problem 1: Build lengths 7, 10, and 14 beside the page. Find every word for each length.

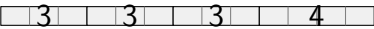
7

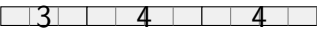
10

14

Problem 2: Find every 3-and-4 word that builds 18. Find a rule that uses word counts for shorter lengths to count the words for any new length.

Use only the sealed box: four 3-rods and three 4-rods. Keep the rods you use and the rods left over in two separate lanes. Do not borrow more rods.

Used  13

Unused  11

Problem 3: With this box, make lengths 6, 9, 10, 19, and 22, or show why a length cannot be made. For each successful build, what length do the unused rods make?

Target	Used rods	Unused rods and length
6		
9		
10		
19		
22		

Problem 4: The whole box has length 24. What rule links the targets that can be made and the lengths left over? Divide all seven rods into two rows of equal length.

For the next problems, you may use as many copies of each stated rod length as you need. Set the sealed box aside.

Problem 5: Test three separate sets of rod lengths: 3, 5, 8; 3, 5, 7; and 3, 5, 4. For each set, find all targets it can make that 3 and 5 cannot, or explain why there are no new targets.

Added rod	New targets	Reason
8		
7		
4		

Problem 6: Build 16 and 24 with the fewest rods. For each target, try once with lengths 3 and 5, and once with lengths 3, 5, and 8. Can a new rod change the fewest-rod answer without adding any reachable targets?

Target	3 and 5	3, 5, and 8
16		
24		
